LANDAU–SIEGEL ZEROS AND A STEPANOV-TYPE NONVANISHING LEMMA

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Abstract. Let χ be a primitive real character of conductor $q \geq 3$. We prove Barreto's bound

$$\sum_{\substack{p \leq X \\ \chi(p)=-1}} \frac{\log p}{p} \leq \frac{1}{2}\log X + \frac{1}{4}\log q + \log\log X + O(1)$$

uniformly in $X \geq 3$ and q, with an effective implied constant, and hence that every zero $\beta \in (0,1)$ of $L(s,\chi)$ satisfies $(1-\beta)\log q \geq 1 - O(\log\log q/\log q)$, so there is no Landau–Siegel zero. Our proof uses a single congruence for both the divisibility and the nonvanishing of an interpolation determinant. As in OpenAI's proof that such zeros do not exist, the norm of the determinant at conjugate pairs (α, α') of quadratic integers is bounded above by the size of its entries and below by its divisibility by high powers of the odd inert primes p, since conjugation agrees with the pth power map modulo p. Modulo a large inert prime p_0, every point lies on the graph $w = z^{p_0}$ of Frobenius, and as z^{p_0} has derivative zero modulo p_0, a linear relation among the rows would give a polynomial in z with more roots, counted with multiplicity, than its degree. This is the multiplicity count of Stepanov's method. The points form Barreto's rectangular box, which gives the $\frac{1}{4}$. A final section compares the proofs of OpenAI, Kerr, Green, Barreto and van Dijk. This paper's mathematics and text were produced by AI systems.

1. Introduction

A Landau–Siegel zero is a real zero β of a quadratic Dirichlet L-function $L(s,\chi)$ with $(1-\beta)\log q$ small, q being the conductor. Other Dirichlet L-functions have none [4, Ch. 14]. Excluding such zeros gives the effective bound $L(1,\chi) \gg 1/\log q$ in place of Siegel's ineffective $L(1,\chi) \gg_\varepsilon q^{-\varepsilon}$, and so, when χ is odd, an effective bound $\gg \sqrt{q}/\log q$ for the class number of $\mathbb{Q}(\sqrt{-q})$. Such zeros were recently excluded by OpenAI [12], as expounded by Banks [1], and since then by Kerr [9], Green [8], Barreto [2] and van Dijk [15]. All of these proofs rest on divisibility by high powers of the inert primes, and they differ in their points, weights and nonvanishing arguments. This paper proves nonvanishing with the congruence that gives the divisibility, by a Stepanov-type root count at an auxiliary inert prime, and lets the weight grow with X, which makes the bound uniform in X and q. With Barreto's rectangular box of points, the argument gives his bounds.

Let χ be a primitive real Dirichlet character of conductor $q \geq 3$, and let $S_\pm(X)$ denote the sum of $(\log p)/p$ over the primes $p \leq X$ with $\chi(p) = \pm 1$. Implied constants are absolute and effective. We use Mertens' theorem $\sum_{p\leq x}(\log p)/p = \log x + O(1)$ and Chebyshev's bound $\sum_{p\leq x}\log p \ll x$.

Theorem 1 (Barreto [2])**.** *For $X \geq 3$ we have*

$$S_-(X) \leq \frac{1}{2}\log X + \frac{1}{4}\log q + \log\log X + O(1).$$

Corollary 2 (Barreto [2])**.** *Every zero $\beta \in (0,1)$ of $L(s,\chi)$ satisfies*

$$(1-\beta)\log q \geq 1 - O\Big(\frac{\log\log q}{\log q}\Big) \qquad \text{and} \qquad (1-\beta)\log q \gg 1.$$

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A real zero close to 1 would make $\chi(p) = -1$ for most primes in a long range, and so make $S_-(X)$ nearly $\log X$, contrary to Theorem 1. For fixed q, Mertens' theorem and $L(1,\chi) \neq 0$ give $S_-(X) = \frac{1}{2}\log X + O_q(1)$, so the $\frac{1}{2}$ cannot be lowered, and the content of Theorem 1 is its uniformity in q.

The idea. Write χ as the Kronecker symbol $(d/\cdot)$ with d a fundamental discriminant, $|d| = q$, and for $\alpha = x + y\sqrt{d}$ put $\alpha' = x - y\sqrt{d}$. If p is odd and $\chi(p) = -1$, so that p is inert in $\mathbb{Q}(\sqrt{d})$, then conjugation on $\mathbb{Z}[\sqrt{d}]/p$ is the Frobenius, so at every point $(z,w) = (\alpha,\alpha')$

$$w \equiv z^p \pmod p, \qquad \text{and} \qquad \partial_z^\kappa (z^p - w) \equiv 0 \pmod p \quad (\kappa \geq 1). \tag{1}$$

The plan is to build a determinant δ with entries of controlled size that is a nonzero algebraic integer divisible by high powers of many inert primes. A nonzero integer has absolute value at least the product of the prime powers dividing it, so the divisibility bounds the norm of δ from below, Hadamard's inequality bounds it from above, and comparing the two gives an upper bound for S_-.

Fix an inert prime p_0 and a weight ℓ, which keeps the w-degrees small (they add to the size of the entries but not to the divisibility, which comes from the z-degrees), and form the matrix with a row for each monomial $z^i w^j$ with $i + \ell j \leq p_0 - 1$, and a column for each point (α,α'), with α in the box $0 \leq x < N_1$, $0 \leq y < N_2$, and each order of z-derivative up to the largest w-degree, about p_0/ℓ, as the root count below requires. Each entry is that derivative of the monomial at the point. Take δ to be a minor that uses every row, and δ' its conjugate, so that $\delta\delta'$ is an integer. There are about $p_0^2/(2\ell)$ rows but only about p_0 points in the box, and the derivatives supply the missing columns without enlarging the box of points.

Divisibility. At an inert prime p with $\ell < p < p_0$, write $i = ph + v$ with $0 \leq v < p$, and replace the row of $z^i w^j$ by that of $z^v (z^p - w)^h w^j$. The rewriting trades z-degree for w-degree. The terms other than $z^i w^j$ have smaller z-degree and, as $p > \ell$, smaller weight, so they are rows of smaller i and δ is unchanged. By (1) and Leibniz's rule, the new row is divisible by p^h, and δ by about $p^{I/p}$, where $I = \sum i$.

Nonvanishing. At p_0 every monomial has z-degree below p_0, so the rewriting does not apply. Instead, (1) is used a second time. Modulo p_0 the points lie on the graph $w = z^{p_0}$ of Frobenius, and since $\partial_z(z^{p_0} - w) \equiv 0$, differentiating in z commutes with substituting $w = z^{p_0}$. Let t be the number of orders of derivative, one more than the largest w-degree. A vanishing combination of rows modulo p_0 would be a polynomial F with $\deg_z F < p_0$ and $\deg_w F < t$ whose z-derivatives of orders $0, \dots, t-1$ vanish at p_0 distinct points of the graph. Then $F(z, z^{p_0})$ would be a nonzero polynomial of degree below $p_0 t$ vanishing to order t at p_0 points, which is impossible by counting roots with multiplicity. So δ can be chosen nonzero. By the second congruence in (1), the derivatives do not weaken the divisibility at the inert primes $p \in (\ell, p_0)$, and at p_0 they supply the multiplicities in the root count. This argument uses characteristic p_0. Over $\mathbb{C}$, the substitution $w = g(z)$ commutes with ∂_z only for constant g, and each line $w = c$ contains at most one of the points. Modulo p_0 the choice $g = z^{p_0}$ is also available, and its graph contains all of them. The count also uses that z^{p_0} has degree only p_0, which keeps $F(z, z^{p_0})$ of degree below $p_0 t$.

Size. Call $\max(|\alpha|, |\alpha'|)$ the height of the point (α,α'). It controls the entries of δ and δ', and the root count requires p_0 points, so we choose the box to minimise their height. A box with sides N_1, N_2 has $N_1 N_2$ points, each of height at most $N_1 + N_2\sqrt{q}$. The best shape balances the two terms: $N_1 \approx p_0^{1/2} q^{1/4}$ and $N_2 \approx p_0^{1/2} q^{-1/4}$, with height $H \approx p_0^{1/2} q^{1/4}$. The points stay distinct modulo p_0 when $q \leq p_0^2$. In that range, up to constants, no set of p_0 points of $\mathbb{Z}[\sqrt{d}]$ does better, since $\#\{\alpha : \max(|\alpha|, |\alpha'|) \leq H\} \ll (1 + H)(1 + H/\sqrt{q})$.

Heuristically, Hadamard's inequality gives $\log|\delta\delta'| \lesssim 2I \log H$, and the divisibility gives $\log|\delta\delta'| \gtrsim 2I \sum(\log p)/p$ over the inert $p \in (\ell, p_0)$, so

$$\sum_{\substack{\ell<p<p_0 \\ \chi(p)=-1}} \frac{\log p}{p} \lesssim \log H \approx \frac{1}{2}\log p_0 + \frac{1}{4}\log q.$$

The right side is at most $\log p_0$, the bound given by Mertens' theorem, exactly when $q \le p_0^2$. Taking p_0 to be the largest inert prime up to X, Mertens' theorem alone gives Theorem 1 for larger q.

In the proof, the w-degrees and the derivatives add $O(\log p_0/\ell)$ to this bound, and the primes $p \le \ell$, to which the rewriting does not apply, add $\log \ell$. We take $\ell \approx \log p_0$. Without derivatives, about $p_0^2/(2\ell)$ points would be needed, of height about $p_0 q^{1/4}/\sqrt{\ell}$. Adding $\log \ell$, this would give only $S_- \lesssim \log p_0 + \frac{1}{4}\log q + \frac{1}{2}\log \ell$, worse than Mertens' bound. The $\frac{1}{4}$ gives the constant 1 in Corollary 2 (Remark 6).

Why inert primes. Inertness enters only through the first congruence in (1), since the second holds for every p. At an odd split prime, Frobenius is the identity on $\mathbb{Z}[\sqrt{d}]/p \cong \mathbb{F}_p \times \mathbb{F}_p$, while conjugation swaps the two factors, so no polynomial congruence $w \equiv g(z)$ holds. The congruences $z^p \equiv z$ and $w^p \equiv w$ that do hold are of little use. The first does not survive differentiation, since $z^p - z$ has z-derivative -1. The second involves only the w-degrees, and the weight ℓ keeps their total below $1/\ell$ times the total z-degree. The argument therefore bounds S_- alone, which is the sum that a Landau–Siegel zero makes large.

2. Two lemmas

Let $\mathcal{O} = \mathbb{Z}[\sqrt{d}] = \mathbb{Z} \oplus \mathbb{Z}\sqrt{d}$.

Lemma 3. *Let p be an odd prime with $\chi(p) = -1$. Then $\mathcal{O}/p\mathcal{O}$ is a field, and $\alpha^p \equiv \alpha' \pmod{p\mathcal{O}}$ for every $\alpha \in \mathcal{O}$.*

Proof. As $(d/p) = \chi(p) = -1$, d is not a square modulo p, so $\mathcal{O}/p\mathcal{O} \cong \mathbb{F}_p[T]/(T^2 - d)$ is a field. By the binomial theorem, Fermat's little theorem and Euler's criterion, $(x+y\sqrt{d})^p \equiv x + yd^{(p-1)/2}\sqrt{d} \equiv x - y\sqrt{d} \pmod{p\mathcal{O}}$. (Oddness is needed: $\mathcal{O}$ need not be maximal at 2.) □

Lemma 4 (after Stepanov [13])**.** *Let k be a field of characteristic p, let $1 \le t \le p$, and let $F \in k[z, w]$ have $\deg_z F < p$ and $\deg_w F < t$. If $(\partial_z^\kappa F)(a, a^p) = 0$ for $\kappa < t$ at p distinct $a \in k$, then $F = 0$.*

Proof. Suppose $F \ne 0$ and put $G(z) = F(z, z^p)$. The monomial $z^i w^j$ becomes z^{i+pj}, and these exponents are distinct because $0 \le i < p$, so $G \ne 0$ and $\deg G \le (p-1) + p(t-1) = pt - 1$. As the derivative of z^p vanishes in characteristic p, the chain rule gives $G^{(\kappa)}(z) = (\partial_z^\kappa F)(z, z^p)$, so $G^{(\kappa)}(a) = 0$ for $\kappa < t$ at each of the p points. Writing $G = \sum_\kappa c_\kappa (z-a)^\kappa$ gives $G^{(\kappa)}(a) = \kappa!\, c_\kappa$, and $\kappa! \ne 0$ in k for $\kappa < t \le p$, so $(z-a)^t$ divides G for each of the p points. The $(z-a)^t$ are pairwise coprime, contradicting $\deg G < pt$. □

Lemma 4 is the multiplicity count of Stepanov's method. Stepanov constructs a nonzero such F vanishing to high order at a set of points and uses the count to bound the size of the set. Here the points are given, and the count shows that $F = 0$, so the matrix M of §3 has full row rank modulo p_0. The count is exact on the rectangle $i < p$, $j < t$: the exponents $i + pj$ fill $0, \dots, pt - 1$ once each, so by Hermite interpolation the pt values $(\partial_z^\kappa F)(a, a^p)$ can be arbitrary. In §3 the lemma is applied with $p = p_0$ to the triangle $i + \ell j \le D$ instead, which, unlike the rectangle, is preserved by the change of basis at each inert prime in $(\ell, D]$.

3. Proof of Theorem 1

Proposition 5. *Let $p_0 \geq 5$ be a prime with $\chi(p_0) = -1$ and $q \leq p_0^2$, put $D = p_0 - 1$, and let $\ell \geq 2$ be an integer. Then*

$$S_-(D) \leq \Big(\frac{1}{2} + \frac{7}{2\ell}\Big) \log D + \frac{\ell+1}{4\ell} \log q + \log \ell + O(1).$$

Deduction of Theorem 1. Let p_0 be the largest prime $\leq X$ with $\chi(p_0) = -1$. If there is no such prime, or $p_0 \leq 3$, then $S_-(X) = O(1)$. Otherwise $S_-(X) = S_-(p_0)$ by maximality. If $q > p_0^2$, then Mertens' theorem gives $S_-(X) \leq \log p_0 + O(1) \leq \frac{1}{2} \log X + \frac{1}{4} \log q + O(1)$. If $q \leq p_0^2$, then $S_-(X) = S_-(D) + O(1)$, and we apply Proposition 5 with $\ell = \lceil \log p_0 \rceil \geq 2$. As $\log D \leq \ell$ and $\log q \leq 2\ell$, the terms $7 \log D/(2\ell)$ and $\log q/(4\ell)$ are $O(1)$, and $\log \ell \leq \log \log X + 1$. □

Proof of Proposition 5. Monomials. Let $s = \lfloor D/\ell \rfloor$, and let $\mathcal{M}$ be the set of $(i, j) \in \mathbb{Z}_{\geq 0}^2$ with $i + \ell j \leq D$, so that $i \leq D < p_0$ and $j \leq s$. Put $m = |\mathcal{M}|$, $I = \sum_{\mathcal{M}} i$ and $J = \sum_{\mathcal{M}} j$. Reflecting $i \mapsto D - \ell j - i$ for each fixed j gives $2I + \ell J = Dm$. Writing $i = \ell u + r$ with $0 \leq r < \ell$, for each r the pairs (u, j) fill a triangle $u + j \leq \lfloor (D - r)/\ell \rfloor$, symmetric in u and j, so $\sum u = \sum j$ and $I \geq \ell J$. Therefore, as $Dm = 2I + \ell J \leq 3I$,

$$J \leq I/\ell, \qquad m \leq 3I/D. \tag{2}$$

A nonzero algebraic integer. Let $N_1 = \lceil p_0^{1/2} q^{1/4} \rceil$ and $N_2 = \lceil p_0^{1/2} q^{-1/4} \rceil$. Then $N_1 N_2 \geq p_0$. Moreover, $N_1 \leq p_0$, because $p_0^{1/2} q^{1/4} \leq p_0$ when $q \leq p_0^2$, and $N_2 \leq \lceil p_0^{1/2} \rceil \leq p_0$. Let M be the matrix over $\mathcal{O}$ with rows indexed by $\mathcal{M}$, ordered by increasing i, columns indexed by (x, y, κ) with $0 \leq x < N_1$, $0 \leq y < N_2$ and $0 \leq \kappa \leq s$, and entries

$$M_{(i,j),(x,y,\kappa)} = \partial_z^\kappa \big(z^i w^j\big)(\alpha, \alpha') = (i)_\kappa \, \alpha^{i-\kappa} \alpha'^j \quad (\kappa \leq i), \qquad \alpha = x + y\sqrt{d},$$

and 0 if $\kappa > i$, where $(i)_\kappa = i(i-1)\cdots(i-\kappa+1)$. Its rows are linearly independent modulo p_0. Indeed, a relation among them over the field $\mathcal{O}/p_0\mathcal{O}$ is a polynomial $F = \sum_{\mathcal{M}} c_{ij} z^i w^j$ with $(\partial_z^\kappa F)(\bar{\alpha}, \bar{\alpha}^{p_0}) = 0$ for $\kappa \leq s$ (Lemma 3) at the $N_1 N_2 \geq p_0$ reductions $\bar{\alpha}$, which are distinct because $N_1, N_2 \leq p_0$ and $1, \sqrt{d}$ are linearly independent over $\mathbb{F}_{p_0}$. As $\deg_z F \leq D < p_0$, $\deg_w F \leq s$ and $s + 1 \leq p_0$, Lemma 4 with $t = s + 1$ gives $F = 0$. Hence we may fix an $m \times m$ minor $\delta \notin p_0\mathcal{O}$ of M, which does not depend on the primes p below. Let δ' be its image under $\sqrt{d} \mapsto -\sqrt{d}$, which is the same minor with α and α' interchanged. Then $n = |\delta\delta'|$ is a positive integer, as $\delta\delta' \in \mathbb{Z} \setminus \{0\}$. (We pass to the norm because $|\delta|$ itself can be small when $d > 0$.)

Its size. Put $H = N_1 + N_2\sqrt{q}$. As $|\alpha|, |\alpha'| \leq x + y\sqrt{q} < H$ and $(i)_\kappa \leq D^s$, every entry in row (i, j) of the matrices of δ and δ' has modulus at most $D^s H^{i+j}$, and Hadamard's inequality gives

$$\log n \leq 2(I + J) \log H + 2ms \log D + m \log m. \tag{3}$$

Its divisibility. Let p be a prime with $\ell < p \leq D$ and $\chi(p) = -1$. As $\ell \geq 2$, p is odd, so Lemma 3 applies. For $(i, j) \in \mathcal{M}$ write $i = ph + v$ with $0 \leq v < p$, and replace the row of $z^i w^j$ by that of

$$z^v (z^p - w)^h w^j = z^i w^j + \sum_{\nu=1}^{h} (-1)^\nu \binom{h}{\nu} z^{i - p\nu} w^{j+\nu}.$$

The monomials in the sum have smaller z-degree and smaller weight $i + \ell j - (p - \ell)\nu$, so they lie in $\mathcal{M}$. Doing this for every row adds to each row integer multiples of rows of smaller i, a unitriangular change, so δ, which uses every row, is unchanged. By Leibniz's rule, $\partial_z^\kappa \big(z^v (z^p - w)^h w^j\big)$ is an integer combination of products each containing h factors $\partial_z^\lambda (z^p - w)$ with $\lambda \geq 0$. Each such factor lies in $p\mathcal{O}$ at (α, α'). For $\lambda = 0$ this is Lemma 3. For $1 \leq \lambda \leq p$ it holds because $\partial_z^\lambda z^p = (p)_\lambda z^{p-\lambda}$ and

$p \mid (p)_\lambda$, and for $\lambda > p$ the factor is zero. So the new row of (i,j) lies in $p^h\mathcal{O}$, hence $\delta \in p^{E_p}\mathcal{O}$. Conjugating, $\delta' \in p^{E_p}\mathcal{O}$, so $p^{2E_p} \mid n$ with

$$E_p = \sum_{(i,j)\in\mathcal{M}} \Big\lfloor \frac{i}{p} \Big\rfloor \geq \frac{I}{p} - m. \tag{4}$$

Comparison. As the pairwise coprime p^{2E_p} all divide n, multiplying (4) by $2\log p$, summing over p and using Chebyshev's bound $\sum_{p\leq D} \log p \ll D$ give

$$2I \sum_{\substack{\ell<p\leq D \\ \chi(p)=-1}} \frac{\log p}{p} \leq 2\sum_p E_p \log p + O(mD) \leq \log n + O(mD).$$

Insert (3) and divide by $2I$. By (2) and $m \leq (D+1)^2$, we have $(I+J)/I \leq 1 + 1/\ell$, $ms/I \leq 3/\ell$, and $(m\log m)/I$, mD/I are $O(1)$. Mertens' theorem gives $\log\ell + O(1)$ for the primes $p \leq \ell$, so

$$S_-(D) \leq \log H + \frac{\log H + 3\log D}{\ell} + \log\ell + O(1). \tag{5}$$

Finally, $N_1 \leq p_0^{1/2}q^{1/4} + 1$ and $N_2\sqrt{q} \leq p_0^{1/2}q^{1/4} + \sqrt{q}$, and $1 \leq \sqrt{q} \leq p_0^{1/2}q^{1/4}$ as $q \leq p_0^2$. Hence $H \leq 4p_0^{1/2}q^{1/4}$ and $\log H \leq \frac{1}{2}\log D + \frac{1}{4}\log q + O(1)$, and the proposition follows. □

4. Proof of Corollary 2

A zero β near 1 makes $-L'/L(\sigma,\chi)$ very negative just to the right of 1, while Theorem 1 bounds it below. We track the constants, as Barreto [2] does, and as Banks [1] does for the OpenAI argument.

Proof of Corollary 2. Lower bound for $-L'/L$. Let $T(x) = \sum_{p\leq x} \chi(p)(\log p)/p = S_+(x) - S_-(x)$, so that $T(x) = 0$ for $1 \leq x < 2$. The primes with $\chi(p) = 0$ divide q, and

$$\sum_{p\mid q} \frac{\log p}{p} \leq \sum_{p\leq \log q} \frac{\log p}{p} + \frac{1}{\log q}\sum_{p\mid q} \log p \leq \log\log q + O(1).$$

So Mertens' theorem gives $T(x) \geq \log x - 2S_-(x) - O(\log\log q)$. Writing $\log^+ = \max(\log, 0)$, Theorem 1 then gives some $K = \frac{1}{2}\log q + O(\log\log q)$ with $K \geq \frac{1}{2}\log q$ and

$$T(x) \geq -K - 2\log^+\log x \qquad (x \geq 1), \tag{6}$$

which is trivial for $x < 3$, where $T(x) \geq -\frac{1}{2}\log 2 > -K$. Let $0 < u \leq 1$ and $\sigma = 1 + u$. The logarithmic derivative is an average of T over logarithmic scales. By partial summation, and since the prime powers contribute $O(1)$ (as $\sum_p (\log p)/(p(p-1))$ converges),

$$-\frac{L'}{L}(\sigma,\chi) = u\int_1^\infty T(x)\, x^{-u-1}\, dx + O(1).$$

The weight $u\,x^{-u-1}$ has total mass 1 and satisfies $u^2\int_1^\infty (\log x)\, x^{-u-1}\, dx = 1$. Since $\log y \leq y$, we have $\log^+\log x \leq \log(1/u) + u\log x$, and so (6) gives

$$-\frac{L'}{L}(\sigma,\chi) \geq -\frac{1}{2}\log q - 2\log\frac{1}{u} - O(\log\log q). \tag{7}$$

Upper bound for $-L'/L$. Let $L(\beta,\chi) = 0$ with $0 < \beta < 1$. Taking real parts in the formula for L'/L that comes from the Hadamard product [4, Ch. 12, (17)–(18), and Ch. 14, p. 89, (2)],

$$-\frac{L'}{L}(\sigma,\chi) = \frac{1}{2}\log\frac{q}{\pi} + \frac{1}{2}\frac{\Gamma'}{\Gamma}\Big(\frac{\sigma+\mathfrak{a}}{2}\Big) - \sum_\rho \operatorname{Re}\frac{1}{\sigma-\rho}, \qquad \chi(-1) = (-1)^{\mathfrak{a}},\ \mathfrak{a}\in\{0,1\},$$

where ρ runs over the nontrivial zeros of $L(s,\chi)$, counted with multiplicity. The gamma term is bounded, as $(\sigma+\mathfrak{a})/2 \in (\frac{1}{2},\frac{3}{2}]$, and every term of the sum is positive because $\operatorname{Re}\rho \le 1 < \sigma$. Keeping only $\rho=\beta$ and using (7),

$$\frac{1}{u+1-\beta} \le \log q + 2\log\frac{1}{u} + O(\log\log q) \qquad (0<u\le 1). \tag{8}$$

Conclusion. Put $\mathcal{L}=\log q$ and $u=\mathcal{L}^{-2}$, so that $u\mathcal{L}=1/\mathcal{L}$ and $\log(1/u)=2\log\mathcal{L}$. By (8),

$$(1-\beta)\mathcal{L} \ge \frac{\mathcal{L}}{\mathcal{L}+O(\log\mathcal{L})} - \frac{1}{\mathcal{L}} \ge 1 - O\Big(\frac{\log\mathcal{L}}{\mathcal{L}}\Big).$$

This bound is empty for small q. To get $(1-\beta)\log q \gg 1$ for every q, take $u=\eta/\mathcal{L}$ instead, with $0<\eta\le 1$. Since $\log(1/u)=\log(1/\eta)+\log\mathcal{L}\le\log(1/\eta)+\mathcal{L}$, (8) gives $1/(u+1-\beta)\ll(1+\log(1/\eta))\mathcal{L}$. Hence $(1-\beta)\mathcal{L}\ge c_0/(1+\log(1/\eta))-\eta$ with $c_0>0$ absolute, and this is positive once η is small enough. □

Remark 6. The constant 1 in Corollary 2 is $1/(\frac{1}{2}+2\cdot\frac{1}{4})$: the $\frac{1}{2}$ comes from the conductor in the Hadamard formula, and $2\cdot\frac{1}{4}$ from Theorem 1, as S_- appears twice in $T=(S_++S_-)-2S_-$. The constant also depends on the coefficient of $\log X$ in Theorem 1 being exactly $\frac{1}{2}$, so that $2S_-$ cancels the $\log x$ of Mertens' theorem. With $\frac{1}{2}+\varepsilon$ in its place, the right side of (8) would gain a term $2\varepsilon/u$, and the argument would give a constant below 1.

5. Relation to other proofs

The OpenAI proof [12], expounded by Banks [1], works over $\mathbb{Q}(\sqrt{d_0},\sqrt{2})$, where d_0 is the squarefree part of d. As both note, it is an interpolation-determinant argument [10], with the smallness p-adic rather than archimedean, as in Dobrowolski's work on Lehmer's problem [5] (a connection noted by K. Pratt in the comments to [14]) and in its determinant form due to Cantor and Straus [3].

Divisibility. All the proofs use the congruence $w\equiv z^p \pmod p$. Green [8] works in $\mathbb{Q}(\sqrt{d})$ with the points (α,α'), takes z-derivatives and applies Leibniz's rule, and orders the rows by z-degree. We follow him in this. Where his main proof uses a Gram determinant, we use a nonzero minor, as a remark in [8, version 2, §2] also suggests. The change of basis to $z^v(z^p-w)^h$ comes from [12], in Kerr's form [9], and is used in truncated form by Green [8, Lemma 5.1] and van Dijk [15].

Points and weights. Green, and van Dijk in his version 1, use a square box and a rectangle of monomials. OpenAI [12] and Kerr [9] give the monomials a fixed weight. Here the weight grows with X, which keeps the losses to $\log\log X$, and Barreto's [2] weight grows too. The box is Barreto's rectangle, which van Dijk's version 2 also uses.

Nonvanishing. OpenAI [12] uses an interpolation lemma with a greedy choice of monomial rows. Kerr [9] constructs an auxiliary polynomial by the pigeonhole principle in place of a determinant, and Masser's zero estimate [11] shows that it cannot vanish at all the points. Green [8, Lemma 3.1] bounds multiplicities by resultants at any points with distinct second coordinates satisfying an incidence bound, which Schwartz–Zippel supplies. Barreto [2] uses a Wronskian and a minor in $\mathbb{Z}$. Van Dijk's note [15] of 20 October proves nonvanishing by the root count of Lemma 4. In his version 1, p_0 is the largest inert prime up to q^k with $k\to\infty$ along hypothetical zeros, the points have height about $\sqrt{p_0 q}$, and as $\log q = o(\log p_0)$ the resulting $\frac{1}{2}\log q$ is negligible in that setting. Here p_0 may be as small as $\sqrt{q}$, and the shape of Barreto's box gives the $\frac{1}{4}$. Tao [14], who expounds Green's argument, has noted in the comments close ties between that approach and Stepanov's version of the polynomial method.

Results. OpenAI, Kerr and van Dijk in his version 1 prove $(1-\beta)\log q\ge c$ with an absolute $c>0$, and Banks proves it with explicit constants. Green proves $S_-(X)\le(\frac{1}{2}+O(1/C))\log X+O_C(1)$ at $X=q^C$ in his version 2, which implies the same. Barreto, and van Dijk in his version 2, prove Theorem 1 and Corollary 2.

Versions. Version 1 of this paper [6], dated 18 October 2026, had the argument of §§2–3 with a square box: the derivatives and the nonzero minor after Green, the weighted monomials and the change of basis after Kerr, a weight growing with X, and the nonvanishing of Lemma 4. It proved Theorem 1 with $\frac{1}{2}\log q$ in place of $\frac{1}{4}\log q$, and Corollary 2 as $(1-\beta)\log q \gg 1$. Version 2 [7] took Barreto's rectangular box and his constant in Corollary 2.

Authorship

The mathematics and text were produced by the AI systems named above, building on their version 1 [6] and on [1, 2, 8, 9, 12], and checked in separate AI reviews. It has not been refereed. Jacob Pescador initiated the project and coordinated the systems and their reviews. He contributed no mathematics, has not verified the proofs, and maintains this posting.

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